

Modern Operating Systems
Chapter 4 – Example File
Systems

Zhang Yang
Spring 2025



Content of the Lecture

- 4.5 Example File Systems



Example File Systems

- The MS-DOS File System
- UNIX V7 File System

Example File Systems Overview

- Many file systems, sometimes many within an operating system.
 - Each with its own format (CD-ROM is ISO 9660; Unix has **UFS**, FFS; Windows has FAT, FAT32, NTFS as well as floppy, CD, DVD Blu-ray, Linux has more than 40 types, with **extended file system** ext2 and ext3 leading; plus distributed file systems, etc.)
 - New ones still arriving – ZFS, GoogleFS, Oracle ASM, FUSE.



The MS-DOS File System (1)

- The first IBM PC used.
- Main file system of Windows 98 and Windows ME.
- Still supported on Windows 2000, Windows XP, and Windows Vista.
- MS-DOS file system and its extension (FAT-32) have become widely used for many embedded systems.



The MS-DOS File System (2)

- Use 8+3 file names.
- DOS 1.0 limited to one directory.
- Later versions allows hierarchical file system.
 - Directory can nested to an arbitrary depth.
- Directories are variable sized.
- But directory entries are fixed size of 32 bytes.

The MS-DOS File System (3)

■ Directory Entry

- Attributes: indicate a file is read-only, archived, hidden, or system file.
- Time: hours (5 bits), minutes (6 bits), seconds (5 bits)
- Date: year (7 bits), month (4 bits), day (5 bits)
- Size: in theory can be 4GB, but in fact 2GB or less.

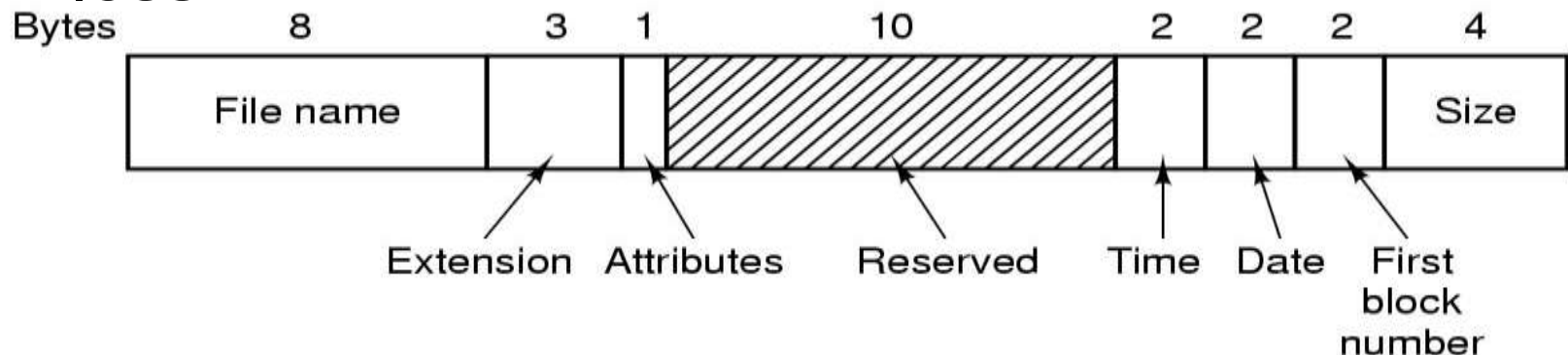


Figure 4-30 The MS-DOS directory entry



The MS-DOS File System (4)

- The highest expressible year in MS-DOS is 2107.
 - The counter starts at 1980.
- Use file allocation table to keep track of file addresses.
- Each directory entry only gives the first block number.
 - Which is used to index into the FAT table.



The MS-DOS File System (5)

- FAT file system has three versions for MS-DOS: FAT-12, FAT-16, FAT-32.
 - Each differs from the bits a disk address contains.
 - FAT-32 introduced with 2nd release of Windows 95.
- Disk blocks are multiples of 512 bytes.
 - Can be different for each partition.
- Allowable block size differing by variants.

The MS-DOS File System (6)

■ FAT-32

- 28-bit disk address
- Partition size: theoretically max 8TB ($2^{28} \times 2^{15}$ bytes), but usually limited to 2TB.

Block size	FAT-12	FAT-16	FAT-32
0.5 KB	2 MB		
1 KB	4 MB		
2 KB	8 MB	128 MB	
4 KB	16 MB	256 MB	1 TB
8 KB		512 MB	2 TB
16 KB		1024 MB	2 TB
32 KB		2048 MB	2 TB

Figure 4-31 Maximum partition for different block sizes. The empty boxes represent forbidden combinations

The MS-DOS File System (7)

■ Advantages of FAT-32 over FAT-16

- Support larger disks.
- An 8-GB disk using FAT-32 can be a single partition. Using FAT-16 has to be four partitions.
- For a given size disk partition, a smaller block size can be used.

The MS-DOS File System (8)

■ Exercise

- Problem: The MS-DOS FAT-16 table contains 64K entries. Suppose that one of the bits had been needed for some other purpose and that the table contained exactly 32768 entries instead. With no other changes, what would the largest MS-DOS file have been under this condition?
- Solution: The largest block is 32KB. With 32,768 of these blocks, the biggest file would be 1 GB.

The UNIX V7 File System (1)

- Introduced in PDP-11.
- Organized in the form of a tree starting from root.
- File names are up to 14 characters and can contain any ASCII characters except / and NUL.
- Each directory entry has two fields:
 - File name, and the # of i-node for that file (2 bytes)
- Number of files in the file system limited to 64K.

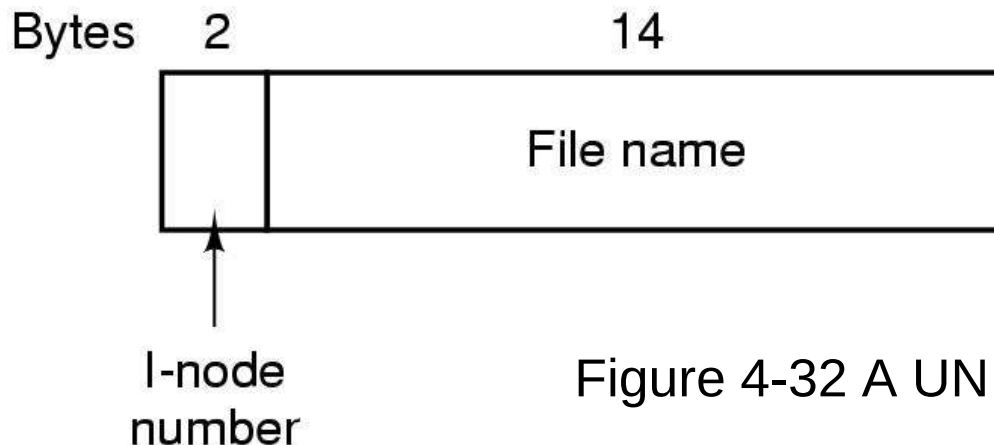


Figure 4-32 A UNIX V7 directory entry

The UNIX V7 File System (2)

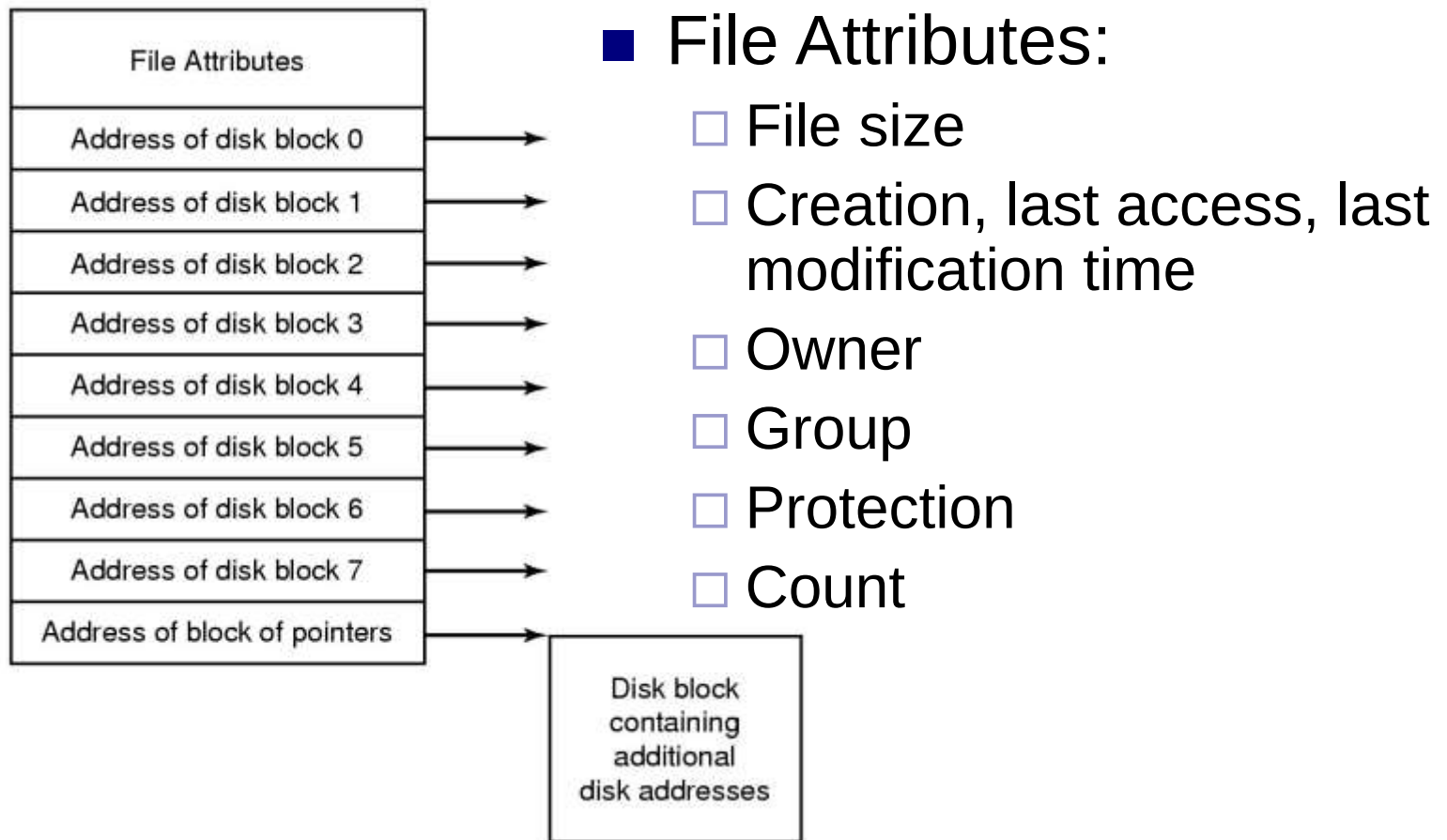


Figure 4-13 An Example i-node

The UNIX V7 File System (3)

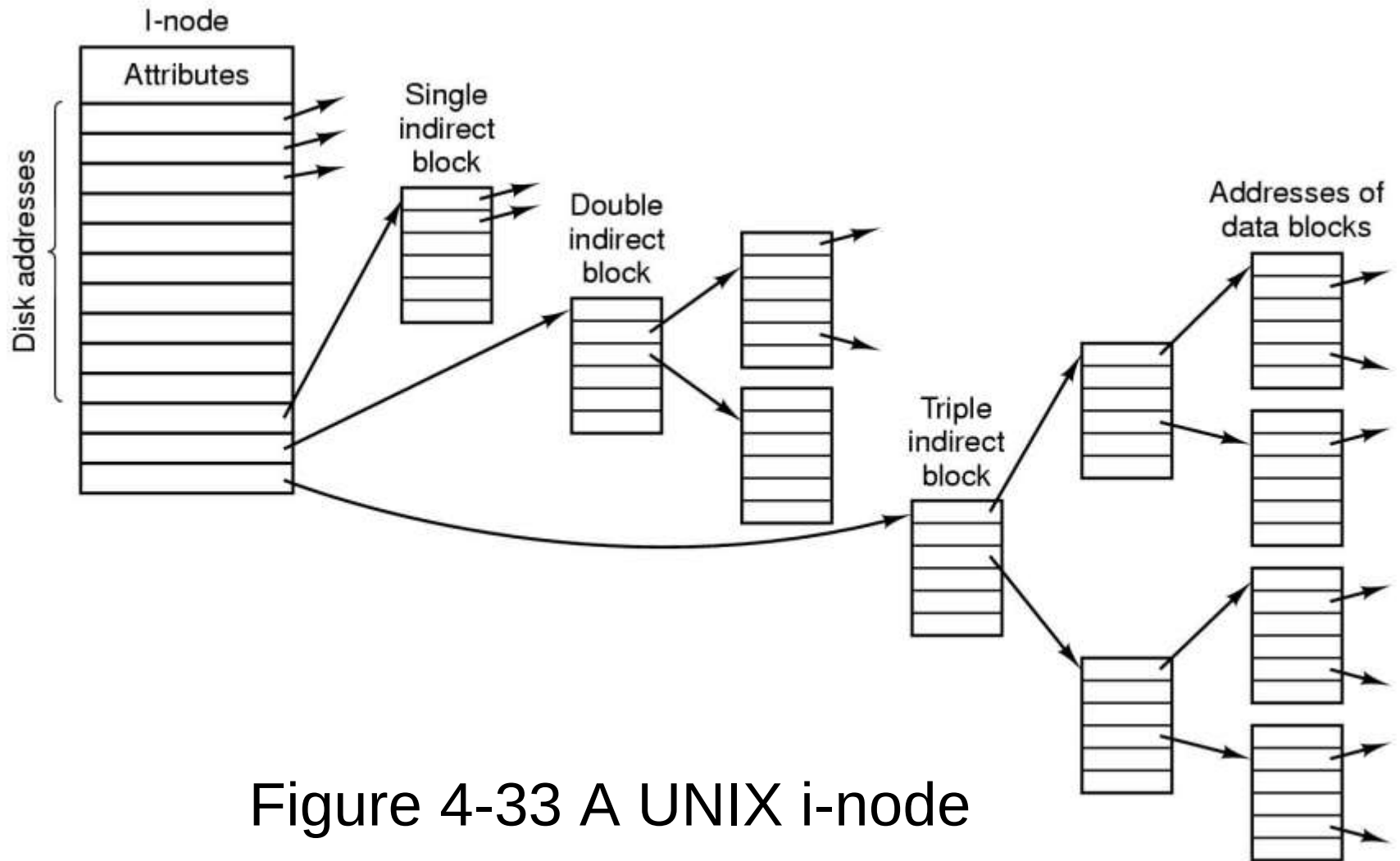


Figure 4-33 A UNIX i-node

The UNIX V7 File System (4)

■ Exercise

- Problem1: A Unix file system has 1KB blocks and 4 byte disk address. What is the maximum file size if i-nodes contain 10 direct entries, and one single, double and triple indirect entry each?
- Solution: The i-node holds 10 pointers. The single indirect block holds 256 pointers. The double indirect block is good for 256^2 pointers. The triple indirect block is good for 256^3 pointers. Adding these up, we get a maximum file size of 16,843,018 blocks, which is about 16.06 GB.

The UNIX V7 File System (5)

- The steps in looking up */usr/ast/mbox*:
 - The file system locates the root directory. It can be anywhere on the disk, but its i-node is located at a fixed place, but assume block 1.
 - The file system looks up the first component of the path (*usr*) and find its i-node number. From this i-node, the system locates the directory for */usr* and looks up the next component (*ast*) in it.
 - When the entry for *ast* is found in directory */usr*, i-node for the directory */usr/ast* can be obtained. From this i-node it can find the directory itself and lookup *mbox*.
 - The i-node for this file is then read into memory and kept there until the file is closed.

The UNIX V7 File System (6)

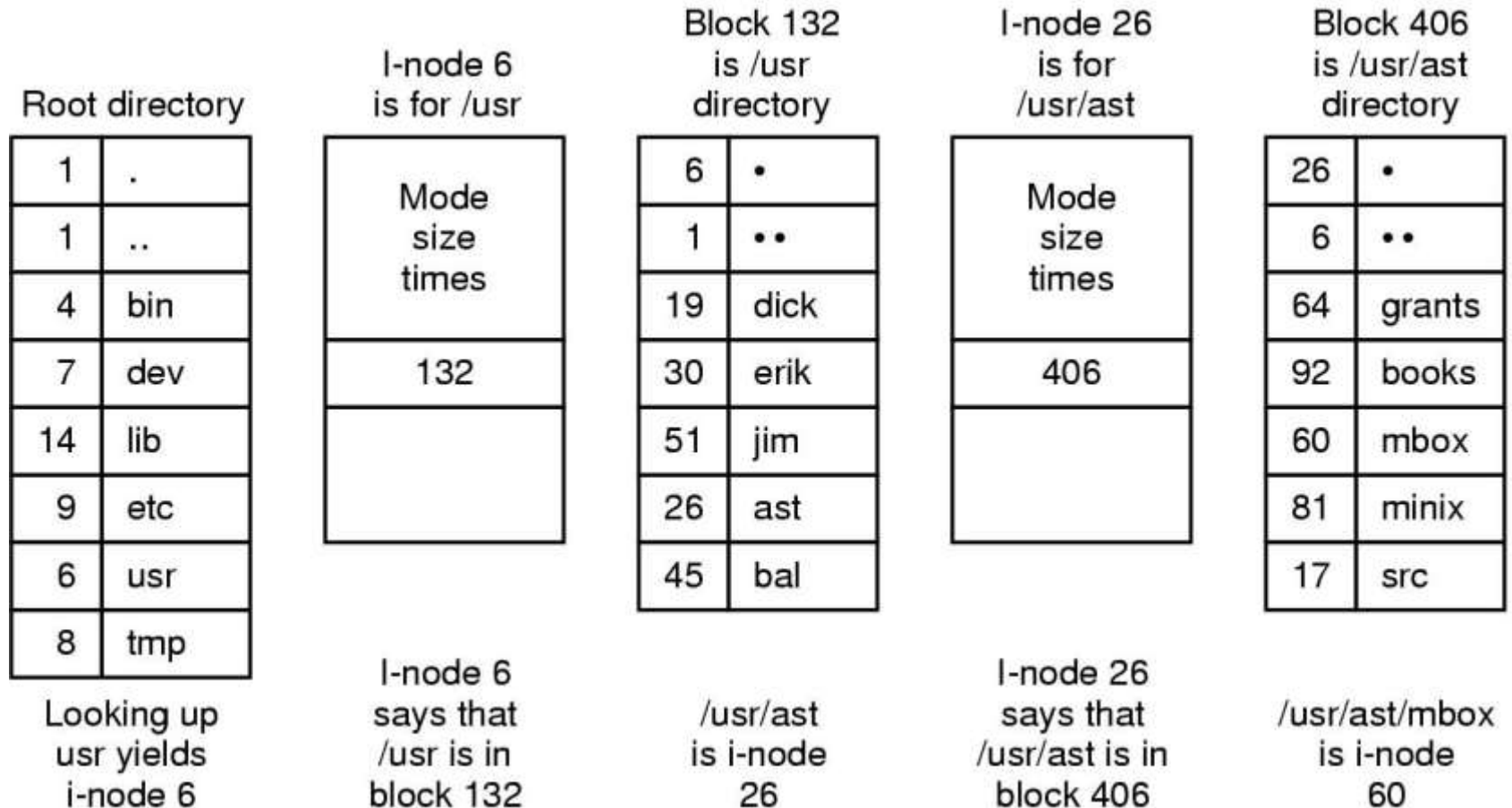


Figure 4-35 The steps in looking up `/usr/ast/mbox`

The UNIX V7 File System (7)

■ Exercise

- Problem 2: How many disk operations are needed to fetch the i-node for the file `/usr/ast/course/os/handout.t`? Assume that the i-node for the root directory is in memory, but nothing else along the path is in memory. Also assume that all directories fit in one disk block.

The UNIX V7 File System (8)

■ Exercise

□ Solution: The following disk reads are needed:

- directory for */*
- i-node for */usr*
- directory for */usr*
- i-node for */usr/ast*
- directory for */usr/ast*
- i-node for */usr/ast/courses*
- directory for */usr/ast/courses*
- i-node for */usr/ast/courses/os*
- directory for */usr/ast/courses/os*
- i-node for */usr/ast/courses/os/handout.t*

In total, 10 disk reads are required.



Homework

- P335 35, 38, 41, 42